

AWS CLOUD PRACTITIONER STUDY GUIDE

Let's Get Started

This Cheat Sheet is your comprehensive study companion for the AWS Cloud Practitioner exam.

Unlike the Cram Sheet (designed for a quick final revision), this guide provides in-depth explanations, detailed service comparisons, and practical examples to help you build a solid understanding of AWS fundamentals. Use this throughout your study journey to dive deeper into each domain, compare similar services, and reinforce your knowledge with real-world analogies.



IDENTITY & ACCESS MANAGEMENT (IAM)

○ IAM USERS VS ROLES:

- Users have long-term credentials tied to one person.
- Roles are temporary and designed to be assumed by apps, services, or users needing temporary access.

○ IAM POLICIES:

- JSON documents that define permissions.
- Apply least privilege — only grant what's necessary.

○ IAM GROUPS:

- Collections of users to manage permissions more easily.

○ MFA (MULTI-FACTOR AUTHENTICATION):

- Always enable MFA on the root account.
- Supports hardware devices, virtual apps, and U2F security keys.

○ IAM CREDENTIAL REPORT:

- Lists all users and the status of their credentials (e.g. password age, active keys).

○ BEST PRACTICES:

- Rotate access keys regularly.
- Do not embed credentials in code – use IAM roles for EC2/Lambda.
- Follow least privilege principle.

COMPUTE SERVICES

○ EC2 (ELASTIC COMPUTE CLOUD):

- **On-Demand:** flexible, pay-as-you-go.
- **Reserved:** 1–3 years, big discount (largest with All Upfront, Standard).
- **Spot:** spare capacity, up to 90% cheaper, but can be interrupted.
- **Dedicated Hosts:** physical servers for compliance.

○ AUTO SCALING:

- Adds/removes instances automatically based on demand.
- Improves availability and lowers cost.

○ ELASTIC LOAD BALANCING (ELB):

- Distributes traffic across multiple targets (EC2, containers, IPs).
- Improves fault tolerance.

○ AWS FARGATE:

- Serverless compute for containers (ECS/EKS).

○ AWS LAMBDA (SERVERLESS):

- Runs code without managing servers.
- Pay per request + compute time.
- Supports Node.js, Python, Java, Go, .NET, and custom runtimes.

○ AWS ELASTIC BEANSTALK:

- Easiest way to deploy apps without managing infrastructure.
- Supports multiple languages/platforms.

○ AWS BATCH:

- Run large-scale batch computing jobs.
- AWS Batch runs batch computing jobs at any scale. It automatically provisions compute resources based on the volume and requirements of jobs.

NETWORKING & CONTENT DELIVERY

○ REGIONS & AVAILABILITY ZONES (AZS):

- Region = geographic area.
- AZ = isolated data centers within a Region.
- Deploy across multiple AZs for high availability.

○ EDGE LOCATIONS:

- Used by CloudFront and Route 53 to deliver content with low latency.

○ AMAZON VPC (VIRTUAL PRIVATE CLOUD):

- Lets you define your own isolated virtual network.
- You control subnets, route tables, gateways, and security settings.

○ SECURITY GROUPS VS NACLs:

- Security Groups = stateful, applied at instance level.
- NACLs = stateless, applied at subnet level.

AMAZON ROUTE 53:

- AWS DNS service.
- Supports routing policies (latency, geolocation, failover, weighted).

AWS GLOBAL ACCELERATOR:

- Improves global application availability and performance by routing traffic via AWS global network.

AMAZON CLOUDFRONT:

- Content Delivery Network (CDN).
- Caches content at edge locations to reduce latency.

STORAGE SERVICES

AMAZON S3:

- Object storage with virtually unlimited scalability.
- Storage classes: Standard, Standard-IA, Intelligent-Tiering, Glacier, Glacier Deep Archive.
- Features: versioning, lifecycle policies, encryption, access control.

AMAZON EBS (ELASTIC BLOCK STORE):

- Block storage for EC2.
- Automatically replicated within AZ.
- Snapshots stored in S3.

AMAZON EFS (ELASTIC FILE SYSTEM):

- Shared file storage, scalable, accessible by multiple EC2 instances.

AMAZON GLACIER / GLACIER DEEP ARCHIVE:

- Long-term archival storage.
- Cheap, but retrieval times vary (minutes to hours).

AWS STORAGE GATEWAY:

- Hybrid storage to integrate on-premises with AWS.

○ AWS SNOWBALL / SNOWMOBILE:

- Physical appliances to transfer petabytes of data to AWS.

AWS STORAGE SERVICES COMPARISON

Type	AWS Services	Analogy	Best For	Key Features
Object Storage	Amazon S3	Giant filing cabinet with infinite drawers	Storing unstructured data like images, videos, backups, logs	Virtually unlimited scale, global namespace, 11 nines durability, lifecycle rules, multiple storage classes
Block Storage	Amazon EBS	USB hard drive attached to a computer	OS disks, transactional databases, applications that need low-latency block-level storage	Persistent storage per EC2 instance, snapshots, high performance
File Storage	Amazon EFS, Amazon FSx	Shared network drive / specialised file cabinets	Shared file access across multiple servers, Windows apps, HPC workloads	Scales elastically, supports NFS/SMB, multi-AZ, managed file systems
Archive Storage	Amazon S3 Glacier, S3 Glacier Deep Archive	Locked basement archive	Long-term backup, compliance, rarely accessed data	Cheapest storage option, retrieval times range from minutes to hours
Hybrid Storage	AWS Storage Gateway	Bridge between office and AWS	Enterprises connecting on-prem storage to the cloud	File/volume/tape gateway modes, cached or stored volumes, seamless hybrid integration
Backup & Centralised Protection	AWS Backup	Office archivist managing all backups	Centralised backup across AWS services (EBS, RDS, DynamoDB, EFS, FSx)	Policy-based backup, compliance reports, cross-region backups

DATABASES

○ AMAZON RDS (RELATIONAL DATABASE SERVICE):

- Managed service for MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Aurora.
- Supports Multi-AZ (disaster recovery) and Read Replicas (scaling).

○ AMAZON DYNAMODB:

- Fully managed NoSQL database.
- Serverless, scales automatically, millisecond latency.

○ AMAZON REDSHIFT:

- Data warehouse for analytics.

AMAZON ELASTICACHE:

- In-memory caching service (Redis, Memcached).
- Improves application performance.

AMAZON NEPTUNE:

- Graph database.

DATABASE SERVICES COMPARISON

Service	Type	Analogy	Best For	Key Features
Amazon RDS	Relational (SQL)	Filing cabinets with drawers	Traditional apps, e-commerce, banking, ERP	Managed MySQL, PostgreSQL, Oracle, SQL Server; automated backups, patching, Multi-AZ
Amazon Aurora	Relational (SQL)	Supercharged filing cabinet	High-performance relational apps	MySQL/PostgreSQL-compatible, up to 5× faster, auto-scaling storage, serverless option
DynamoDB	NoSQL (Key-Value)	Notebooks on your desk	Gaming, mobile apps, shopping carts, IoT	Serverless, auto-scaling, millisecond latency, pay-per-request billing
ElastiCache	In-Memory (Redis/Memcached)	Sticky notes	Caching, real-time analytics, session storage	Sub-millisecond latency, reduces load on primary DB, supports Redis & Memcached
MemoryDB	In-Memory (Redis)	Sticky notes with better glue	Durable Redis-compatible workloads	Redis-compatible, high availability, persistence + durability
Neptune	Graph Database	Whiteboard with connections	Social networks, fraud detection, recommendation engines	Stores relationships as first-class objects, supports Gremlin & SPARQL
DocumentDB	Document Store	Folders full of varied files	JSON-based apps, content management, catalogs	MongoDB-compatible, flexible JSON storage, fully managed
Timestream	Time-Series	Wall calendar tracking events	IoT sensors, metrics, financial/time-series data	Built for time-series queries, serverless, automatic scaling
DMS (Migration)	Migration Tool	Moving truck	Moving DBs into AWS	Migrates with minimal downtime, supports heterogeneous migrations
SCT (Conversion)	Schema Conversion	Organising files for new cabinet	Adapting DB schemas for AWS engines	Converts schema/code from Oracle/SQL Server to Aurora/MySQL/PostgreSQL

MONITORING & LOGGING

AWS CONFIG:

- Tracks resource configurations and compliance over time.

○ **AMAZON CLOUDWATCH:**

- Monitors metrics (CPU, memory, network).
- Alarms can trigger SNS notifications or Auto Scaling.

○ **AWS CLOUDTRAIL:**

- Logs all API calls in the account.
- Key for security auditing.

○ **AWS X-RAY:**

- Helps debug and analyse applications, trace requests.
- AWS X-Ray helps developers analyse and debug distributed applications by tracing requests across services and identifying performance bottlenecks.

○ **TRUSTED ADVISOR:**

- Provides recommendations in categories:
Cost, Performance, Security, Fault Tolerance, Service Limits.

SECURITY & COMPLIANCE

○ **SHARED RESPONSIBILITY MODEL:**

- AWS secures the cloud (infrastructure, hardware, global network).
- Customer secures data, OS, apps, IAM.

○ **AWS KMS (KEY MANAGEMENT SERVICE):**

- Manage encryption keys.

○ **AWS CLOUDHSM:**

- Dedicated hardware-based key storage.

○ **AMAZON GUARDDUTY:**

- Threat detection service.

○ **AMAZON INSPECTOR:**

- Automated security assessment service.

○ **AWS ARTIFACT:**

- Download AWS compliance reports and certifications.

○ **AWS WAF (WEB APPLICATION FIREWALL):**

- Protects apps from common exploits (SQL injection, XSS).

BILLING & PRICING

○ **PRICING MODELS:**

- Pay-as-you-go.
- Save with Reserved Instances or Savings Plans.
- Spot Instances for cheap, interruptible workloads.

○ **COST EXPLORER:**

- Visualise and analyse costs.

○ **AWS BUDGETS:**

- Set custom budgets with alerts.

○ **AWS COST & USAGE REPORT:**

- Most detailed billing data.

○ **TCO CALCULATOR:**

- Compare AWS costs with on-premises.

○ **CONSOLIDATED BILLING (AWS ORGANISATIONS):**

- Combine multiple accounts for volume discounts.

GENERAL AWS CONCEPTS

WELL-ARCHITECTED FRAMEWORK PILLARS:

- Operational Excellence.
- Security.
- Reliability.
- Performance Efficiency.
- Cost Optimisation.

CLOUD BENEFITS:

- Pay-as-you-go pricing.
- Agility and speed.
- Elasticity and scalability.
- No upfront hardware cost.
- Global reach with Regions and AZs.

AWS SERVICE COMBINATIONS

LESSON 4.9

WEB APP



STATIC SITE



SERVERLESS BACKEND



GOVERNANCE



COMPARISON OF OTHER AWS SERVICES

Category	Service	What It Does (Function)	Analogy	Use Case Example
Application Integration	Amazon SQS	Managed message queue to decouple apps	Post office (delivers mail reliably)	Order processing, job queues
	Amazon SNS	Pub/sub notifications to many subscribers	Town crier / loudspeaker	Send alerts, fan-out messages
	Amazon EventBridge	Event bus for routing events between apps	City events coordinator	Automate workflows, trigger Lambda
Business Apps	Amazon Connect	Cloud contact center service	Call center at city hall	Customer support, call routing
	Amazon SES	Email sending service	City's official postal system (email)	Marketing campaigns, notifications
Developer Tools	CodeCommit	Source code repo (Git)	Filing cabinet for code	Version control
	CodeBuild	Build & test service	Factory machine assembling products	CI builds, testing
	CodeDeploy	Automated deployments	Delivery truck	Push updates to servers
	CodePipeline	CI/CD orchestration	Conveyor belt assembly line	End-to-end app delivery
	X-Ray	Tracing & debugging service	Mechanic's diagnostic tool	Troubleshoot microservices
	Cloud9	Cloud IDE	Shared co-working office	Team coding
End-User Computing	WorkSpaces	Virtual desktops	Virtual office desks	Remote workforce
	AppStream 2.0	App streaming to users' devices	Streaming office apps	Software trials, remote apps
Frontend & Mobile	Amplify	Build & deploy full-stack apps quickly	Construction kit for apps	Mobile/web app dev
	AppSync	Managed GraphQL API service	Restaurant waiter	Deliver exactly the requested data
IoT	IoT Core	Device connection & management	City's traffic control center	Smart homes, IoT sensors
	IoT Greengrass	Local processing for IoT devices	Mini brain in each neighbourhood	Edge analytics
AI & ML Apps	Personalise	Real-time recommendations	Personal shopper	Retail, streaming
	Comprehend	NLP: sentiment, keywords, entities	Language expert	Customer feedback analysis
Customer Engagement	AWS Activate, IQ, Managed Services	Consulting & business support	City planners, consultants	Startup credits, expert help
	AWS Support	Support plans for AWS accounts	City help desk	Troubleshooting, enterprise support

AWS PRICING MODEL COMPARISON CHART

Model	How it Works	Best For	Discount vs On-Demand	Analogy
On-Demand	Pay per use (per sec/hr), no commitment	Unpredictable or dev/test usage	0% (baseline)	Hotel room
Reserved Instance	Commit 1–3 yrs for specific instance	Predictable, steady workloads	Up to 72%	Apartment lease
Savings Plans	Commit to \$ spend, flexible usage	Mixed or variable workloads	Up to 66%	Gym membership
Spot Instance	Use spare capacity, can be reclaimed	Batch jobs, fault-tolerant apps	Up to 90%	Last-minute flight
Dedicated Host	Full physical server	Compliance, licensing needs	N/A	Buying a house
Dedicated Instance	Hardware isolation, not pinned host	Isolation, moderate flexibility	Slight premium	Apartment block
Capacity Reservation	Reserve AZ capacity upfront	Mission-critical availability	0% unless combined	Restaurant reservation

SUPPORT PLANS COMPARISON TABLE

Plan	Analogy	Features
Basic	Owner’s manual only	Docs, forums, whitepapers, Trusted Advisor core checks. No tech support.
Developer	Basic roadside assistance	Email support during business hours. Guidance for dev/test.
Business	Full roadside rescue & towing	24/7 phone/chat/email, faster response, full Trusted Advisor, production use.
Enterprise On-Ramp	Mechanic on call for mid-sized businesses	TAM, proactive reviews, mission-critical guidance.
Enterprise	VIP personal mechanic riding along	TAM, fastest response, concierge-level support, proactive architecture help.

COMPARISON OF KEY TECHNICAL RESOURCES

Resource	Analogy	Purpose
Documentation & Whitepapers	Instruction manuals & textbooks	Official guides and best practices.
Knowledge Center & re:Post	Study groups & Q&A forums	Ask questions, FAQs, troubleshooting help.
Trusted Advisor	Financial & safety advisor	Automated account checks (cost, performance, security, fault tolerance).
Health Dashboard	Doctor’s health chart	Global AWS service status + personalised health events.
Professional Services	Specialist consultants	Expert human help for migrations, scaling, or architecture.
APN Partners	Certified contractors	Vetted businesses to support cloud projects.